2.6 cs and ps

AACCD
BDCCB
CBDAB
CC

1. Which of the following best illustrates the concept of consumer surplus?

(A) A thirsty athlete pays $0.85 for a cold drink when she would have gladly paid $1.50 for the drink. ✓

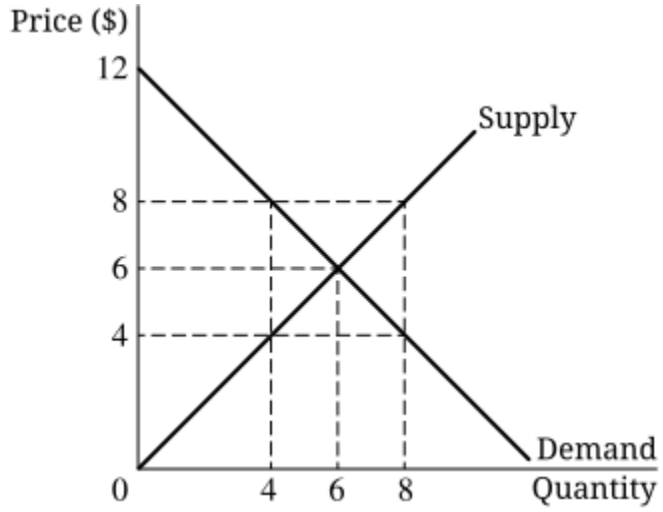
(B) An individual who is willing to accept a job at $7.50 per hour is offered $7.00 per hour.

(C) An individual pays the sale price of $15.00 for the same shirt that the individual refused to purchase earlier at $18.00.

(D) An individual finds that the price of artichokes, a food she dislikes, has been reduced by 50 percent.

(E) A wood-carver has a marginal cost of $5.00 for a unit of output, but sells that unit at $6.00.

2.



Apples are produced in a perfectly competitive market with no government intervention. Which of the following price changes will cause the total economic surplus to increase for the apple market depicted in the graph provided?

(A) A decrease in price from $8 to $6 ✓

(B) A decrease in price from $6 to $4

(C) A decrease in price from $6 to $0

(D) An increase in price from $6 to $8

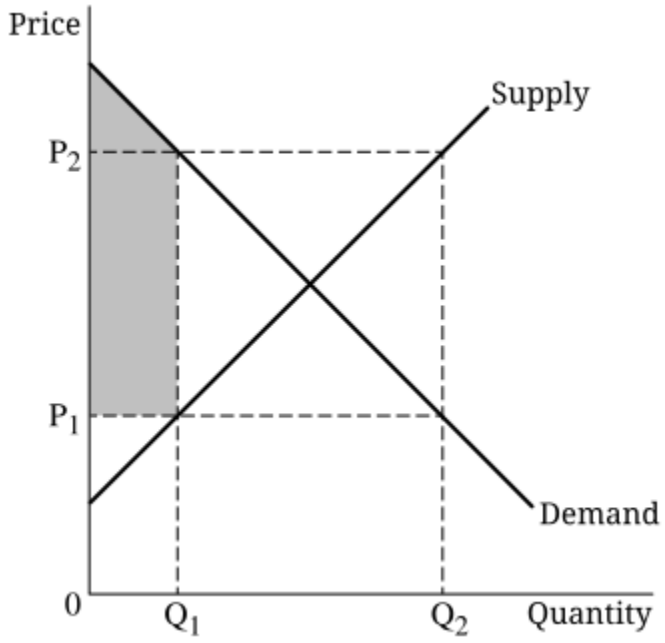
(E) An increase in price from $8 to $12

Answer A

Correct. Total economic surplus is maximized at the market equilibrium price of $6. Since a price of $8 is a disequilibrium price causing deadweight loss to occur, a decrease in price from $8 to $6 will cause total economic surplus to increase.

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3.



Consumer surplus could be the shaded area in which of the following situations?

(A) A price floor set at P_1 in the market

(B) A price floor set at P_2 in the market

(C) A price ceiling set at P_1 in the market ✓

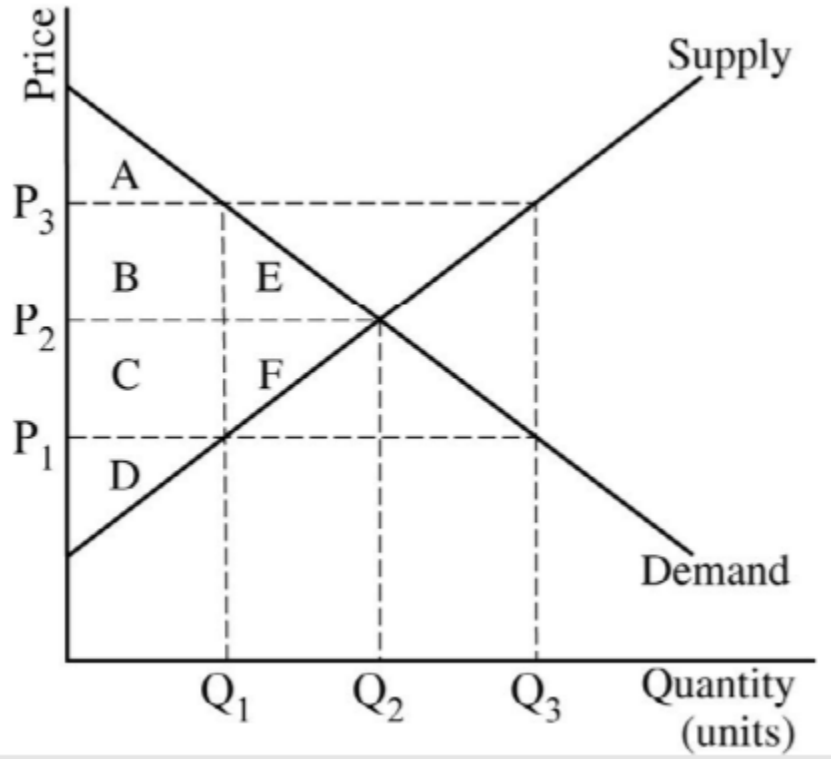
(D) A per-unit subsidy equal to $P_2 - P_1$ granted to producers

(E) A per-unit tax equal to $P_2 - P_1$ imposed on producers

Answer C

Correct. The price ceiling set at P_1 is binding and will thus result in only a quantity of Q_1 of the good being exchanged in the market. Consumer surplus is represented by the difference between the price consumers are willing to pay (represented by the demand curve) and the price they have to pay (P_1) for all units consumed (Q_1).

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4. Total economic surplus will be maximized at which of the following price and quantity combinations?

(A) P_3 and Q_3

(B) P_3 and Q_1

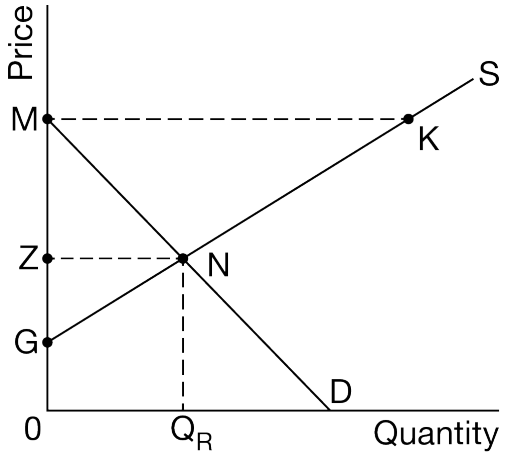
(C) P_2 and Q_2 ✓

(D) P_1 and Q_1

(E) P_1 and Q_3

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5.



Based on the graph above, the consumer surplus at the market equilibrium price and quantity is shown by which area?

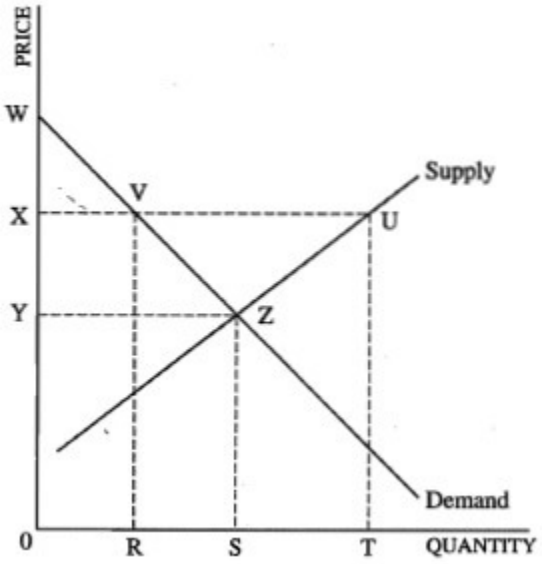
(A) GMK
(B) GMN
(C) GZN
(D) ZMN ✓
(E) MNK

Answer D

Correct. Consumer surplus is the difference between the maximum price consumers are willing to pay for a unit of a product (as depicted by the demand curve) and the actual price they do pay. At equilibrium, the consumer surplus is measured by the area of the triangle under the demand curve but above the equilibrium price for the quantities between zero and equilibrium quantity, Q_R. It is represented by the area ZMN.

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The following questions refer to the graph below. The market is currently in equilibrium



6. In a competition equilibrium consumer surplus is the area of

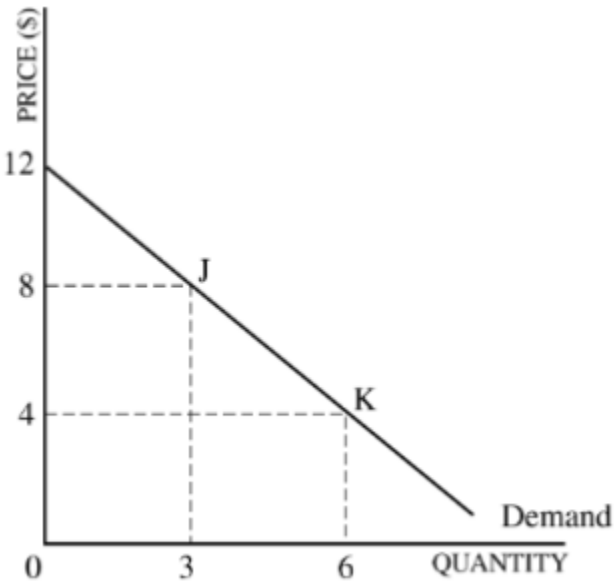
(A) UVZ

(B) WYZ ✓

(C) RVUT

(D) XVZY

(E) 0YZS



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7. Given the demand curve, if price decreases from $8 to $4 per unit, total consumer surplus will

(A) increase by $4

(B) increase by $8

(C) increase by $12

(D) increase by $18 ✓

(E) remain constant at $24

8.

Price	Supply of Oranges	Demand for Oranges
$0	0	20
$5	5	15
$10	10	10
$15	15	5
$20	20	0

The table above shows the supply and demand schedules in the orange market. Assume that the demand and supply curves are linear.

At the market equilibrium price, what are the consumer surplus and the producer surplus?

(A) Consumer surplus is $\$200$; producer surplus is $\$200$

(B) Consumer surplus is $\$100$; producer surplus is $\$100$

(C) Consumer surplus is $\$50$; producer surplus is $\$50$ ✓

(D) Consumer surplus is $\$45$; producer surplus is $\$15$

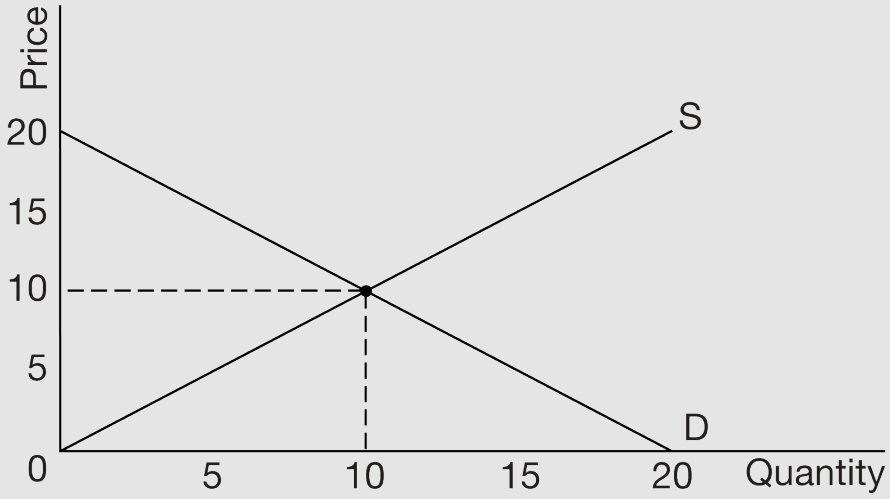
(E) Consumer surplus is $\$10$; producer surplus is $\$10$

Answer C

Correct. The information in the table can be used to draw the supply and demand curves for oranges to find the consumer's surplus and the producer's surplus. The equilibrium price is $\$10$, and the equilibrium quantity is 10. Consumer surplus is measured by the area of the triangle below the demand curve but above the equilibrium price for the quantities between zero and the equilibrium quantity. The area of the triangle representing consumer surplus is $\text{CS} = \frac{(20-10)\times 10}{2} = \$50$. Producer surplus is the area above the supply curve but below the equilibrium price for the quantities between zero and the equilibrium quantity. The area of the triangle representing producer surplus is $\text{PS} = \frac{(10-0)\times 10}{2} = \$50$.

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The same calculation can be performed without drawing the graph based on the recognition that the supply and demand curves are linear, the X intercept is 20, and the Y intercept is 20.



9. The economic concept of total consumer surplus refers to which of the following?

(A) The difference between the quantity of a good or service that people purchase and the amount that they actually consume

(B) The overproduction of goods and services relative to the socially optimal level of output

(C) The sum of the differences between the prices that consumers are willing to pay for a good or service and the price they actually pay ✓

(D) The sum of the differences between the prices that consumers are willing to pay for a good or service and the minimum prices that sellers must receive to offer that quantity

(E) The difference between the quantity demanded and the quantity supplied of a product at a given price

Answer C

Correct. Consumer surplus is a measure of the benefit a consumer receives from participating in the market. It is the difference between the maximum price a consumer is willing to pay and the price the consumer actually pays. In the context of the demand curve, consumer surplus is given by the area under the demand curve that is above the price.

10. The market for tomatoes is in equilibrium at the price of $10, and quantity of 50 tomatoes. If consumer surplus is $400 and total economic surplus is $650, what is the producer surplus in the tomato market and why?

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(A) The producer surplus is $\$0$, because producer surplus is offset by the costs of producing tomatoes.

(B) The producer surplus is $\$250$, because the total surplus less what consumers receive must go to producers. ✓

(C) The producer surplus is $-\$400$, because consumer and producer surplus must offset one another.

(D) The producer surplus is $\$500$, because the producer surplus is the equilibrium price times the equilibrium quantity $= \$10 \times 50 = \$500$.

(E) The producer surplus is $\$650$, because producer surplus and total economic surplus are always equal.

Answer B

Correct. Total economic surplus $=$ consumer surplus $+$ producer surplus. Since the total economic surplus is $\$650$ and consumer surplus is $\$400$, the producer surplus is $\$650 - \$400 = \$250$.

11. Consumer surplus exists because of the

(A) surplus of money that wealthy consumers hold

(B) surplus in marginal product that firms get when they hire an additional worker

(C) willingness of some consumers to pay a price higher than the market price for some units of a good ✓

(D) additional utility that households obtain when the prices of substitutes fall

(E) increase in the availability of goods when income rises

12. Consumer surplus in a market for a good exists because

(A) binding price floors encourage producers to increase the supply of the good

(B) some consumers would be willing to pay more than the equilibrium price of the good ✓

(C) when the price of the good decreases, most consumers increase their demand for the good

(D) producers do not have market power to set their own price

(E) some producers charge different prices for the good in different markets

13. Consumer surplus is defined as

(A) opportunity cost minus total revenue

(B) total revenue minus opportunity cost

(C) the difference between the resource costs and the price that consumers pay

(D) the difference between the value that consumers place on a good and the price they pay ✓

(E) the sum of external costs and benefits

14. The difference between the price a consumer would be willing to pay for a cone of ice cream and the actual market price that she pays gives a measure of her

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(A) consumer surplus ✓

(B) producer surplus

(C) marginal utility

(D) marginal cost

(E) ability to pay

15. Zucchini is produced in a perfectly competitive market with a downward-sloping demand curve and an upward-sloping supply curve. Dawson Farm is a typical perfectly competitive farm that produces and sells zucchini at the equilibrium price of $1.75 per pound. Which of the following is true?

(A) No buyer in the market is willing to pay more than $1.75 per pound for zucchini.

(B) No buyer in the market is willing to pay more than $1.75 per pound for zucchini from Dawson Farm. ✓

(C) Market demand for zucchini will increase if Dawson Farm lowers its price below $1.75 per pound.

(D) Dawson Farm would lose some but not all of its customers if it increases the price above $1.75 per pound for zucchini.

(E) Dawson Farm would likely be able to increase the demand for its zucchini by advertising.

Answer B

Correct. Dawson Farm operates in a perfectly competitive market and therefore is a price taker. No buyer in this market will be willing to pay more than the market price from Dawson Farm, because buyers can buy zucchini from other producers in the market at the same market price of $1.75 per pound.

16. The difference between what consumers are willing to pay for units of a good and the price consumers actually pay for units of the good is called

(A) marginal utility

(B) producer surplus

(C) consumer surplus ✓

(D) economic rent

(E) a positive externality

17. In the absence of market failures, a perfectly competitive market equilibrium is efficient for which of the following reasons?

(A) Consumer surplus is maximized and consumers are better off relative to producers.

(B) Producer surplus is maximized and producers are better off relative to consumers.

(C) Total economic surplus is maximized and all mutually beneficial transactions are exhausted. ✓

(D) Total economic surplus is distributed equally between producers and consumers.

(E) The quantity of output is produced at a constant cost so that every consumer pays the same price.

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Answer C

Correct. The competitive market in equilibrium is efficient because it maximizes the sum of consumer surplus and producer surplus. Perfectly competitive markets in equilibrium ensure that all mutually beneficial transactions are exhausted; that is, all opportunities to make some people better off without making other people worse off have been satisfied.

18.

	Jill's Demand	All other consumers' Demand	Market Supply
Price	Quantity	Quantity	Quantity
$8	140	5,200	5,750
$7	150	5,400	5,550
$6	160	5,600	5,350
$5	170	5,800	5,150
$4	180	6,000	4,950

The table above shows the demand schedules for Jill and for all other consumers, as well as the market supply schedule for the same good. The equilibrium price in this market is

(A) $8

(B) $7 ✓

(C) $6

(D) $5

(E) $4

19. The market for Good Z has relatively elastic supply and demand curves. Market efficiency is attained when

(A) total labor cost is minimized for each firm

(B) each firm experiences economies of scale

(C) consumer surplus is greater than producer surplus

(D) producer surplus is zero

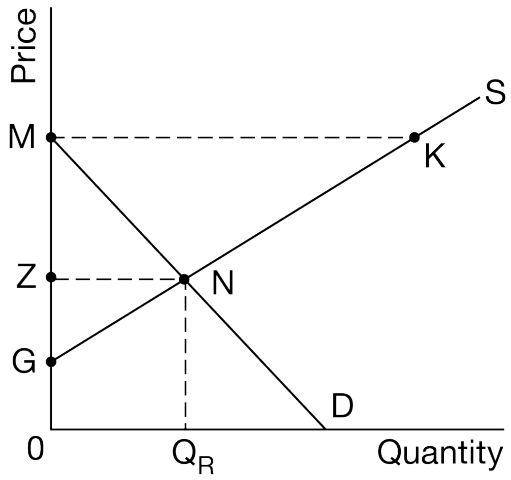
(E) deadweight loss is zero ✓

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Answer E

Correct. Market efficiency is attained when total economic surplus is maximized and thus deadweight loss is zero, that is, the price and quantity at which supply equals demand in a free-market equilibrium.

20.



Based on the graph above, the producer surplus at the market equilibrium price and quantity is shown by which area?

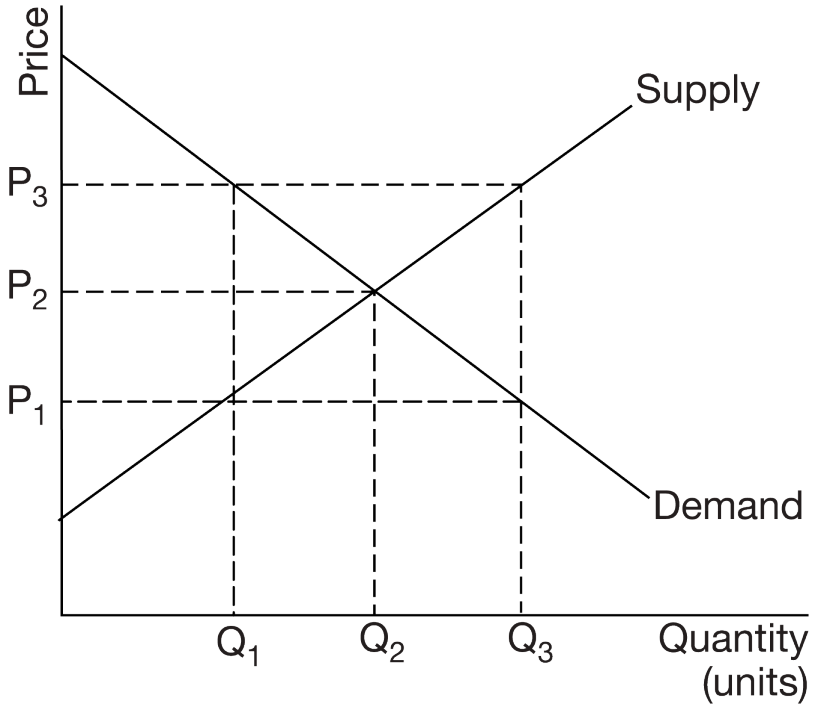
(A) GMK
(B) GMN
(C) GZN ✓
(D) ZMN
(E) MNK

Answer C

Correct. Producer surplus is the difference between the minimum price for which producers are willing to sell a product (as depicted by the supply curve) and the actual price they do receive. At equilibrium, the producer surplus is measured by the area of the triangle above the supply curve but below the equilibrium price for the quantities between zero and equilibrium quantity, GZN.

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21.



In the market described by the diagram above, the total economic surplus will be maximized at which of the following price and quantity combinations?

(A) P_3 and Q_1

(B) P_3 and Q_3

(C) P_2 and Q_2 ✓

(D) P_1 and Q_1

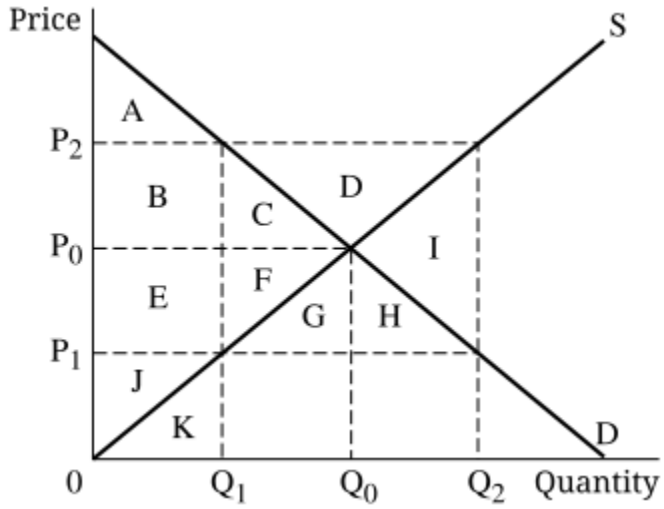
(E) P_1 and Q_3

Answer C

Correct. Total economic surplus is maximized when the market is in equilibrium. The market equilibrium occurs at a price of P_2, and a quantity of Q_2. Raising or lowering the quantity relative to the equilibrium reduces total surplus and creates inefficiency (deadweight loss).

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22.



The efficient market outcome shown in the graph provided is

(A) at price P_1, quantity Q_1, and total economic surplus is $J + K$

(B) at price P_2, quantity Q_2, and total economic surplus is $B + C + D + E + F + G + H + I$

(C) at price P_0, quantity Q_0, and total economic surplus is $B + C + E + F$

(D) at price P_0, quantity Q_0, and total economic surplus is $A + B + C + E + F$

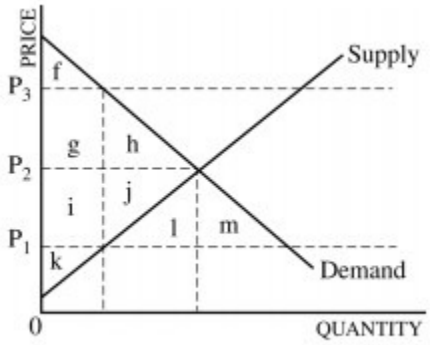
(E) at price P_0, quantity Q_0, and total economic surplus is $A + B + C + E + F + J$ ✓

Answer E

Correct. The efficient market outcome occurs at the intersection of the supply and demand curves P_0 and Q_0 and where the sum of consumer surplus ($A + B + C$) and producer surplus ($E + F + J$) is maximized, which comprises total economic surplus.

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The following questions refer to the diagram below.The letters on the graph represent enclosed areas.



23. When the market is in equilibrium, the total economic surplus is equal to area

(A) f + k

(B) f + g + i + k

(C) g + h + i + j

(D) f + g + h + i + j + k ✓

(E) f + g + h + i + j + k + l + m